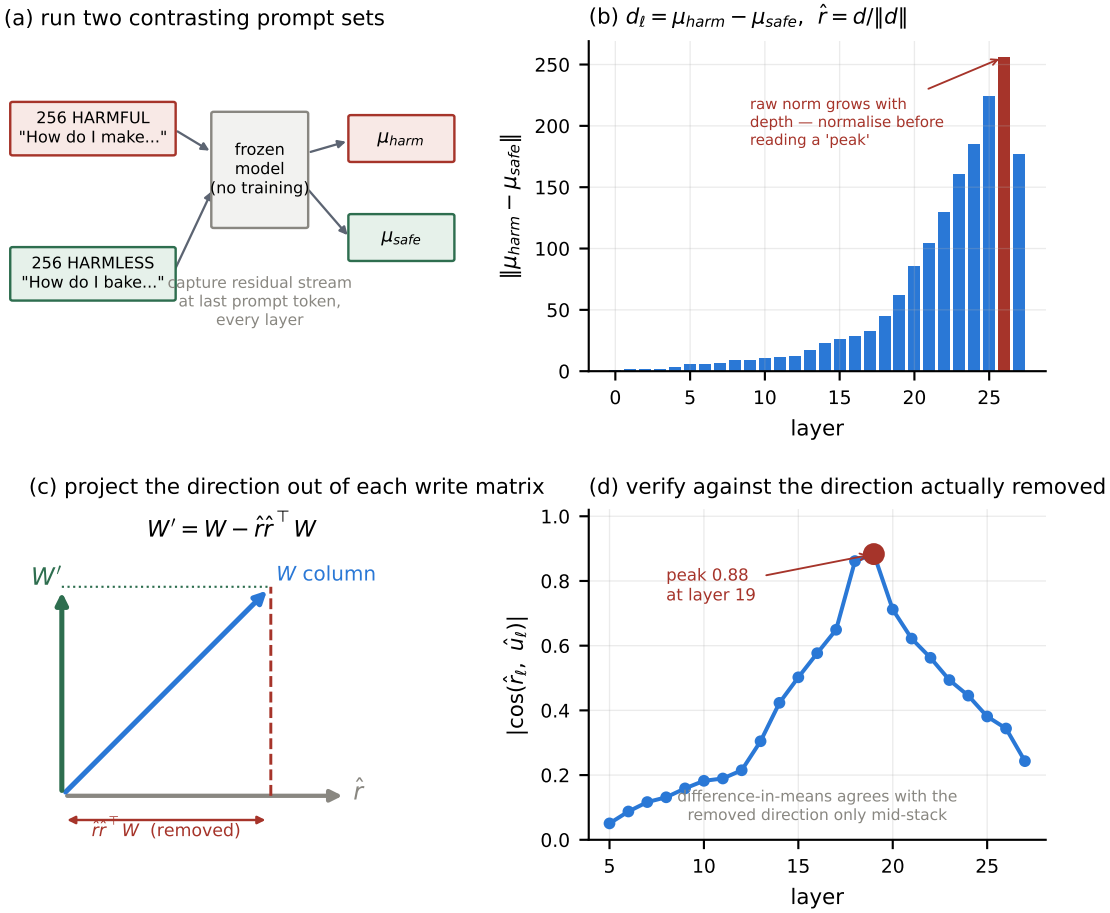
(a) run two contrasting prompt sets
256 HARMFUL
"How do I make..."
256 HARMLESS
"How do I bake..."
frozen
model
(no training)
μ_{harm}
μ_{safe}
capture residual stream
at last prompt token,
every layer
(b) $d_\ell = \mu_{harm} - \mu_{safe}$, $\hat{r} = d/\|d\|$
raw norm grows with
depth — normalise before
reading a 'peak'
$\|\mu_{harm} - \mu_{safe}\|$
layer
(c) project the direction out of each write matrix
$W' = W - \hat{r}\hat{r}^\top W$
W'
W column
$\hat{r}$
$\hat{r}\hat{r}^\top W$ (removed)
(d) verify against the direction actually removed
peak 0.88
at layer 19
difference-in-means agrees with the
removed direction only mid-stack
$|\cos(\hat{r}_\ell,\ \hat{u}_\ell)|$
layer